Addis Ababa University

College of Natural Sciences

Department of Computer Science

Masters Thesis Guideline

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Preface

The Department of Computer Science has been running its masters programme since the 2002/03 academic year. Mostly, the sources of information for students doing theses were their advisors and fellow students. It is believed that lack of information on some issues regarding thesis was an impediment to the academic performance of some students.

Writing a masters thesis is a test of intelligence, endurance and commitment. It opens up the chance to organize and present scholarly work in an intelligent and convincing manner to a wide audience. Theses will be made publicly available, and as widely as possible, in keeping with one of AAU's primary goals of disseminating knowledge.

A thesis must meet the scholarly requirements of the research discipline and be formatted in accordance with the guidelines set out in this document.

This guideline is prepared in the hope that it will bridge some of the information gaps that students experience. It outlines the procedures for preparing a proposal, writing and examination of theses (i.e., from inception to completion of theses and projects), and the roles and responsibilities of students and advisors. It will provide the necessary information for students, advisors, and examiners on rules and procedures pertaining to projects and theses. It is also believed that advisors would benefit from this guideline in fulfilling their advising duties. Moreover, it is a way of encouraging and creating a supportive research environment for students and academic staff.

It is hoped that if masters students know and practice the details in this guideline, they will find it easy to cope up with the rigour of high academic study and make their stay at the Department of Computer Science as fruitful and successful as possible.

The sources of information for this guideline are the Senate Legislation, guidelines of the Office of Graduate Studies as well as practices of the Department. Comments are welcome from readers to improve it in successive revisions. It has also to be noted that some of the requirements as written in this guideline are conventions* that are set to standardize the thesis research process and may vary from institution to institution.

This guideline is divided into six Sections:

- Section 1 provides brief history of the Department.
- Section 2 outlines the differences between a thesis and a project. It also outlines the differences between a basic research and an applied research.
- Section 3 covers details of thesis proposal preparation, submission and defense.
- Section 4 provides some important points in writing a thesis. It includes some typing considerations, how the thesis shall be organized, how tables and figures are used, and on statistical data analysis. It also provides detailed information on the use of references.

* The American Heritage Dictionary defines "Convention" as a general agreement on or acceptance of certain practices or attitudes.

- Section 5 outlines the roles and responsibilities of advisors and students for a smooth conduct of a thesis.
- Section 6 is on thesis submission and defense procedures. It also provides tips on how to prepare slides for the defense and how to present the final work.
- Finally, the annexes contain useful templates regarding thesis proposal, thesis proposal defense result submission, thesis format/organization, and list of items that must be submitted to the department after a successful defense of a thesis.

WE WISH YOU SUCCESS IN YOUR ENDEAVORS

Department of Computer Science

College of Natural Sciences

Addis Ababa University

1. The Department of Computer Science - Brief History

The introduction of computer-related courses at AAU has been informal. Instructions in computer-related subjects started in the Department of Mathematics, in the then Faculty of Science, in the early 1980s without acquiring sufficient computer facilities. To alleviate the ever-increasing demand for professionals and to lay the foundation for future growth and expansion to different types of certification in Computer Science, a Computer Unit was established under the Department of Mathematics in 1986. In the same year, the Unit launched an extension Diploma program in Computer Science with computer facilities acquired from the French Embassy in Addis Ababa. The program helped in producing middle-level professionals in Computer Science that are instrumental to the promotion and advancement of the development endeavors of the country. As a result of the University's shift in policy, the Diploma program terminated as of the 2004/05 academic year.

Realizing the need for high-level professionals in Computer Science, the Unit launched a BSc Degree regular program in 1993 and this was later expanded to the extension division. To further strengthen and expand Computer Science education, the University Senate decided to establish an independent Department of Computer Science in August 2002.

The increasing number of higher education institutions and the shortage of highly qualified computer professionals in the various sectors of the economy demanded the launching of an MSc Degree program in Computer Science in the 2002/03 academic year. The MSc Degree program in Computer Science was launched in the extension division as of the 2010/11 academic year. Since its establishment, the Department has graduated more than 1150 students in the BSc and more than 300 students in the MSc programs (both regular and extension).

2. Differences between a Thesis and a Project

2.1 Thesis vs. Project

One of the difficult questions students raise is: "what is the difference between a thesis and a project?" The distinction between a thesis and a project is not crystal clear. Both a thesis and a project allow students to demonstrate and build their abilities to reflectively apply knowledge and expertise. Both are completed under the guidance of an advisor and carried out in accordance with standards and procedures appropriate to the area of study. In general, they vary in depth and breadth.

The choice also depends on the student's career goals. A thesis might be more appropriate for students who are considering continuing in a PhD program, since the MSc thesis will help them prepare to write a dissertation later, and the MSc topic might easily provide the foundation for a more ambitious PhD research work. A thesis has a large research component as well as a larger, more formal requirement for a written report as compared to a project. A thesis would be appropriate if it can result in a publishable research report.

While a thesis is a scholarly research study (either quantitative or qualitative), a project is more creative in nature. The purpose of a Masters Thesis or Project is to be of educational value to the student, to independently create and present a large and an interesting piece of work. A project is expected to emphasize on implementation while a thesis is expected to emphasize on basic or applied research. With a thesis it is the ideas that are central, while in a project it is what one can develop (e.g., software) that is central.

Project is a significant undertaking appropriate to professional fields. Contributions and results from a project work are compiled into a project report. The report describes or documents the problem, the project's significance, the background analysis performed, the methodologies used, analysis, design, implementation, references to works that were reviewed through the analysis, and lessons learnt in the process in a conclusion or recommendation form.

Thesis is a scholarly treatment of a subject or an investigative treatment of a problem. It identifies the problem, states the major assumptions, explains the significance of the undertaking, sets forth the sources and methods of gathering information, analyzes the data, and offers a conclusion or recommendation. The finished product evidences originality, critical and independent thinking, appropriate organization and format, and thorough documentation.

The purpose of a thesis is to give the student the opportunity to practice what has been learnt in the various courses and ultimately to develop the capability to conduct research independently after completing the program. A thesis requires an extensive literature survey on existing works and has the potential to generate new knowledge or improve upon existing techniques. One of the primary goals of masters-level thesis work is to teach students how to do research.

In a thesis, the student investigates for an original work including a study of its possible implications, its potential applications, and its relationship to previous related works reported in the literature.

In terms of effort, content, and presentation, a thesis goes well beyond the level of a project. The result of a thesis, which is documented as a report, should be able to produce a material that can potentially be acceptable for publication in a journal or a conference proceeding.

In both cases, the student must make a contribution to the existing body of knowledge/experience. A thesis uses an implementation as a proof of concept, but also describes what is conceptually novel and how this work fits into the context of existing works in the field.

A student doing a project is required to produce a system that is functional and usable. Hence, such a student has to carry out a user acceptance testing where the final product is given to potential users and they give their ratings regarding the functionalities of the system and other aspects. The student is also required to produce a users' manual that describes the steps to use the developed system.

2.2 Basic vs. Applied Research

As described above, the core task of a thesis is research. This Section elaborates the differences between basic and applied research.

Research is the systematic study directed towards fuller scientific knowledge of the subject studied. In research, we move from established knowledge to new knowledge. Research is classified into two categories: *basic research* and *applied research*.

Basic research (although known as fundamental research or pure research) is driven by a curiosity or interest in a scientific question. The main motivation is to expand knowledge. The objective is to gain fuller knowledge or understanding of the fundamental aspects of phenomena without specific applications towards a process or product in mind. In other words, there is no obvious commercial value to the discoveries that result from basic research. The benefits are only seen in the long-term.

Applied research refers to a scientific study designed to solve practical problems rather than to acquire knowledge for knowledge's sake. The objective is to use the knowledge or understanding gained from research, directed towards the production of useful materials, devices, systems, or methods. Hence, applied research is used to find solutions to everyday problems and develop innovative technologies.

There is a feeling these days for a shift in emphasis away from purely basic research and toward applied research. This trend is necessitated by the problems faced by many countries as a result of overpopulation, pollution, etc. Many organizations are not also interested to avail financial resources for basic research since they don't envisage any immediate benefits; commercial or otherwise.

However, there are also some who believe that a basic and fundamental understanding of all branches of science is needed in order for progress to take place. In other words, basic research lays down the foundation for the applied science that follows.

Regardless of the controversies and despite the fact that both basic and applied research are required, the emphasis in the Department of Computer Science is on applied research in Computer Science that contributes to solving national problems in agriculture, health, industry, etc. We believe that, although basic research has a scientific merit, developing countries like Ethiopia cannot afford the financial and human resources needed for carrying out basic research. Moreover, world-wide trend shows a shift from basic research to applied research.

Hence, students are encouraged to select thesis topics in applied research.

2.3 Summary

In summary, a project differs from a thesis in one significant way. A thesis presents a research result that contributes to the scholarly literature of Computer Science. A project, in contrast, contributes to the Computer Science profession via implementation of known ideas and theories and/or creative accomplishment. With a project, the student spends the vast bulk of his/her time investigating and implementing solution(s) to real world problems. With a thesis option, the student spends more time on investigation and experimental design and implementation.

3. Thesis Proposal Preparation, Submission, and Defense

From this point onwards in this document, both a thesis and a project are referred to as a "thesis", since the guideline applies to both, the difference being in depth and breadth. When required, we will indicate what differs in the two cases.

A student wishing to do a thesis must first select a topic, write a proposal, and submit the proposal to the Department in two bound copies. The proposal has to be prepared in consultation with a prospective advisor. The prospective advisor can help in many ways. The advisor is better suited to advise the student if a given problem qualifies for a thesis or a project, if the proposal is prepared by incorporating all required components, and most importantly, if the work can be considered as original and the objectives are achievable within the given time frame. The proposal is then defended and finally approved by the Department Graduate Committee (DGC). Details regarding proposal writing, submission and defense are described in the subsequent sections.

3.1 Proposal Writing

The organization of the thesis proposal shall be in adherence to the standard template provided in Annex A - Thesis Proposal Template. The guideline for thesis writing that is presented in Section 4 also applies for writing a proposal. Some of the items to be included in the proposal are described in this Section. Note that the proposals for a thesis and a project have similar structures. The materials to be included are also more or less the same. The variation is only in depth and breadth of coverage and, of course, the magnitude and nature of the problem to be solved.

1. **Introduction**

   This section should provide background and context, clearly identifying the topic area. It should show how the proposed research contributes to knowledge and the production of new products, services, methods or techniques in the subject area.

   Describe the background of the problem, giving a measure of its magnitude (how widespread and important it is).

2. **Motivation**

   Briefly justify what motivates you to do a research under this title.

3. **Statement of the Problem**[*]

This section should show what is already known and the gaps to be filled by the thesis. It should provide a clear and concise description of the central problem to be investigated and the questions to be answered.

4. **Objectives**

Clearly outline the general and specific objectives. State the general objective, describing the ultimate goal of the research, and the specific objectives. The general objective is usually written in a single sentence. The objectives should be specific and realistic in terms of capacity, resources, and time. The specific objectives are to meet the general objective. Avoid using very general and obvious activities as specific objectives. For instance, "Drawing conclusions;" should not be written as a specific objective since every thesis should have a conclusion whether it is stated as an objective or not. During the actual work, it is possible that these objectives may be modified.

- Note: General and specific objectives do not have section numbers and they do not appear in the table of contents page.

5. **Methods**

Usually, this section lists the activities to be carried out in order to achieve the objectives. Describe in general terms the methods to be employed to achieve the objectives of the research including how the data (if any) will be collected, how it will be analyzed, and how evaluation is to be conducted.

* Although there is no strong justification for page limit, the first three sections should not be more than four pages.

Show how each specific objective will be achieved, with enough detail to enable an independent and informed assessment of the proposal.

- Note: It is not advisable to include development tools at this stage since they will be identified later during the research and shall be included in the "Implementation" chapter of the final thesis.

6. Related Work

The proposal should show its position with respect to the literature in the proposed study (connecting the proposed study to the body of knowledge as found from literature). The review of related works should show the student's level of knowledge in the area of his/her proposed topic of research. By reviewing related works, the proposal should show what has been done by others and the gap to be bridged by the proposed work.

- Note: The section title is "Related Work" and not "Related Works".

7. Scope and Limitations

Limitations that may be beyond the control of the researcher and restrict the research's conclusions should be indicated here. The restrictions that may be placed on the research by the student and that may affect conclusions need to be specified. The proposed work could be one that tries to achieve part of a bigger problem. In this case, the scope has to be limited based on time and other resources. The remaining part can be proposed as future work later in writing the thesis which can be done by the researcher himself/herself or by others.

8. Application of Results

Outline the probable application of results and who will benefit from your results and how.

Annexes

Timetable

Provide an estimate of the time needed to carry out the proposed research by indicating each principal phase.

Give a breakdown of the work into its component stages, estimating the amount of time which will be required for the work involved in each of the stages and giving an approximate timetable for the completion of each stage, and the thesis as a whole, including the write up of the thesis.

- Note: The timetable should be given in the form of a Gantt Chart.

Cost

Give an itemized listing of the costs involved in the thesis excluding those costs which are normally borne by the Department for services such as computer facilities and Internet connection. The cost may include such items as stationery materials, photocopying, thesis writing and binding, etc. Special equipment or software that are not available in the Department can also be requested, but requires an agreement by the advisor and a special approval by the DGC. Such materials shall be returned to the Department at the end of the thesis work.

References

Provide a list of references to works cited in the proposal. Refer to Section 4.4 for details about the use of references.

3.2 Thesis Proposal Submission and Defense

A student intending to do a thesis must select a topic and submit a proposal to the Department, approved by his/her advisor, and defend the proposal for a final approval by the DGC. After approval, a student is required to fully engage himself/herself in the research by seeking advice from his/her advisor(s).

A thesis proposal shall be submitted to the Department in two bound copies (one copy for the advisor and one copy for the examiner). The DGC appoints an Examining Committee consisting of the advisor and one academic staff who serves as an examiner. The date and venue of the defense must be posted at least one week prior to the defense date. The proposal must also be distributed to the examiner and the advisor at least one week prior to the defense date.

Then, the student defends his/her proposal before the Examining Committee. The defense session is open to anyone who is interested to attend.

The objective of the proposal defense is mostly to provide feedback to the student. It is intended to clarify the thesis proposed and gain agreement from the committee on the amount and quality of work expected.

The major points that must be considered in evaluating a proposal include, but are not limited to, the following:

- Soundness of hypotheses to be tested or objectives to be achieved (in terms of methods/new technologies to be developed or improved, etc.)
- Originality and innovativeness of the proposed research,
- Relevance/importance of the proposed research,

- Contribution of the proposed research to knowledge,
- Contribution to national goals, priorities, and aspirations,
- Adequacy of the review of related works,
- Demonstration of awareness of previous and alternative approaches to the identified problem,
- Clarity and consistency of objectives/hypotheses with the problem statement,
- Suitability and feasibility of the descriptions of methods,
- Adequacy of time allocated,
- Expected significance of the impact of the results,
- Probability of success of the research within the scheduled time, and
- Format and overall organization as per this guideline.

The decision of the Examining Committee on a proposal may be one of the following:

- *Approved*: The proposal successfully addresses all of the review criteria. No further action is required from the student.
- *Clarification Required*: The proposal meets most of the review criteria but lacks some requirements as decided by the examiner that must be addressed in a revised version. The proposal will not require resubmission in a subsequent review cycle. The advisor and the examiner shall ensure that the indicated revision requirements are met.
- *Revise and Resubmit*: The proposal fails to meet some major criteria as decided by the examiner, but is deemed to be of sufficient potential merit to encourage a resubmission and defense.

- *Rejected*: The proposal fails to meet most or all criteria, and does not have the required scientific merit.

A report of the decision of the Examining Committee is then submitted by the examiner to the DGC using the form attached as Annex B - Thesis Proposal Defense Result Submission Form.

4. Thesis Writing Guideline

A thesis explores one or more research questions or tests a hypothesis. It is important that the thesis be conceptualized and structured.

All writing must be coherent, logical, and easy to follow, keeping the audience in mind. Arguments or positions presented must be clearly supported with facts or citations. Unsubstantiated generalizations must be avoided. Organization is a critical component of clear writing. It helps the reader understand the central idea of what is being written and provides a logical sequence of communication.

The following aspects must be addressed.

- The problem and question(s) being pursued.
- The theoretical base and the literature within which the question(s) has been framed.
- The process or method of investigation.
- Findings or observations. For a project, a description of the purpose of the project, a discussion of the process used in its development, and the final "product" (usually included as an annex) have to be documented. If the project is implemented, a description of the process along with observed results should be included.
- Conclusions, recommendations (if any), and future work.

4.1 Organization of the Thesis

The organization of the thesis shall be in adherence to the standard template provided in Annex C - Thesis Template. A description of some of the items is provided in this Section.

a. Title Page

The title page must be as follows. It has no page number.

Addis Ababa University

College of Natural Sciences

Title of the Thesis

Full name of the student[*]

A Thesis Submitted to the Department of Computer Science in Partial Fulfillment for the Degree of Master of Science in Computer Science

Addis Ababa, Ethiopia

Date (Month Year)

[*] Don't use "By:" before name of the student.

b. Signature Page (approval sheet by the Examining Committee)

The following must be included after the title page. It has no page number. Signatures are required after the defense when the final version of the thesis is prepared.

Addis Ababa University

College of Natural Sciences

*Full name of the student**

Advisor: *Full name of the advisor*

This is to certify that the thesis prepared by *name of student*, titled: *Title of the Thesis*† and submitted in partial fulfillment of the requirements for the Degree of Master of Science in Computer Science complies with the regulations of the University and meets the accepted standards with respect to originality and quality.

Signed by the Examining Committee:

	Name	Signature	Date
Advisor:			
Examiner:			
Examiner:			

* Don't use "By:" before name of the student.

† The title shall be in *italic*.

c. Abstract

An abstract is a brief, comprehensive summary of the contents of the thesis. It gives readers an overview of all the key ideas presented.

The abstract bridges the gap between the title and the main body and thus should be brief and informative. It should broadly summarize the overall content of the thesis, point out new information, and it replaces the need for a summary. It should indicate the major outcomes of the research.

A good abstract is accurate: it reflects exactly what is in the thesis; it is self-contained; it defines any technical terms and avoids using any acronyms or abbreviations; it is concise and specific; and it is brief and to the point. Normally, an abstract is written after the thesis is completed.

The abstract should be restricted to approximately 250 words and should not be more than one page. Keywords of not more than seven should be given following the abstract.

- Note: The abstract has no page number.
- Note: Don't include new ideas in an abstract that were not reported in the thesis.
- Note: Avoid using references in the abstract section.

d. Dedication

A dedication page (if required) can be included.

- Note: Dedication page has no page number.

e. Acknowledgments

Credit must be given to all who have made significant contribution to the research. These may, for example, include instructors, advisors, technicians, information providers, individuals who in one way or another have contributed to the success of the thesis, and organizations (e.g., providers of financial support, facilities, etc.).

- Note: Acknowledgments page has no page number.

f. Table of Contents

Table of contents must be generated automatically and not manually. The title (i.e., Table of Contents) must be centered.

If there are many tables, figures, algorithms, and acronyms/abbreviations, then a "List of Tables", "List of Figures", "List of Algorithms", and "Acronyms/Abbreviations" should be included on separate pages after the Table of Contents page. Similar to Table of Contents, List of Tables, List of Figures, and List of Algorithms must all be automatically generated and the page titles must be centered.

Title page, signature page, abstract, dedication, and acknowledgements must not be included in the Table of Contents.

Page numbering starts (page number i) on the first page of Table of Contents using lower case Roman numerals. Use proper section breaks so that each section can be formatted separately.

Use proper indentation for sub-sections. Also use hanging indent if a title flows to the second line.

g. Acronyms and Abbreviations (if any)

If there are many acronyms and/or abbreviations, it is advisable to have “Acronyms and Abbreviations” page for quick reference. In any case the acronym’s expanded version must be written at first use, which afterwards only the acronym can be used.

An abbreviation and an acronym are both shortened versions of something else*. Both can often be represented as a series of letters. There is an overlap between abbreviations and acronyms. Every acronym is an abbreviation because the acronym is a shortened form of a word or phrase. However, not every abbreviation is an acronym, since some abbreviations - those made from words - are not new words formed from the first few letters of a series of words.

An abbreviation is a shortened form of a word or phrase, as “Mr.” for “Mister”, “Dr.” for "Doctor". When we write the date, we may abbreviate both the day of the week (Mon., Tue., Wed., Thu., Fri., Sat., and Sun.) and the month of the year (Jan., Feb., Aug., Sept., Oct., Nov., Dec.).

An acronym, technically, must spell out another word. We write “RAM” for “Random Access Memory” or “CPU” for “Central Processing Unit”.

- The acronyms/abbreviations have to be written sorted in ascending order.

Write acronyms in uppercase letters. For instance, a temporary storage location within the computer is called *ram*. In this sentence “ram” should be written as “RAM”.

* Source: http://abbreviations.yourdictionary.com/

Don't include known and widely accepted acronyms such as LAN, WAN, CPU, RAM, ROM, etc. in the Abbreviations and Acronyms list. Neither you are required to provide their expanded version at first use.

h. Body of the Thesis

Refer to Section 4.2 for the details.

i. References

It is understood that all of the writing (both wording and ideas) in the thesis is the student's own, unless appropriate attribution is made. Failure to reference properly, even though unintended, constitutes plagiarism, as well as poor scholarship or academic dishonesty. See Section 4.4 for details in the use of references.

- Note: References has no chapter number.

j. Annexes

These include short source program listings and other detailed information that cannot be put within the body of the thesis. Each annex should be titled and listed in the Table of Contents. Each annex must also be referred in the body of the thesis at least once.

- Note: Annexes have no chapter/section numbers. Each Annex has to be numbered as "Annex A", "Annex B", etc.

k. Signed Declaration Sheet

The last page of the thesis must contain a signed declaration by the student and the advisor with the following sentence and the lines that follow it:

> I, the undersigned, declare that this thesis is my original work and has not been presented for a degree in any other

university, and that all source of materials used for the thesis have been duly acknowledged.

Declared by:

Name: ____

Signature: ____________________________________

Date: ____

Confirmed by advisor:

Name: ____

Signature: ____________________________________

Date: ____

4.2 Body of the Thesis

There is no single standard on how a thesis shall be structured. In general it may include the following chapters. Please note the following:

- Each chapter will have sections and sub-sections. Unnecessary breakdown of sections must be avoided, i.e., avoid having section numbers for very short text. If it contains short text (less than a page) it is better to use numbered or bulleted lists. Avoid orphan sub-sections (i.e., there have to be at least two sub-sections within a section).
- Use a final full stop only for chapter numbers such as "1." or "2.". Don't use a full stop at the end of section numbers for second and higher level sections. Write as "2.1" and not "2.1." or as "2.1.1" and not as "2.1.1.".
- Try to avoid too many levels. It is acceptable if there are up to three levels such "2.1.1". If there are more than three levels

(although not recommended) such as “2.1.1.1” or “2.1.1.1.1”, then avoid these from appearing in the table of contents.

- Capitalize the first letter of each word of a chapter title, a section title and paper titles in the “References” section (with the exception of articles, conjunctions, and prepositions).
- It is sometimes observed that there are two entries for the same chapter in table of contents when it is automatically generated such as the one shown below. Use the features of the word processor or the way chapter titles are written to avoid such duplications.

  Chapter 1 1

  Introduction 1
- Don’t use space(s) or tab(s) before section numbers.
- Briefly introduce the chapter at the beginning (of the chapter) by outlining what each section will cover.

a. Introduction

This chapter includes Background (this is the introduction section of the thesis proposal), Motivation, Statement of the Problem, Objectives, Methods, Scope and Limitations, and Application of Results that were part of the thesis proposal but possibly modified in the course of the research. In addition it will also include a last section on Organization of the Rest of the Thesis which should briefly describe the chapters starting from chapter 2 (chapter 1 is already read at this stage and should not be included) and it has to be written in future tense.

The related works section of the proposal shall be expanded and becomes part of Chapters 2 and 3.

- Note: The Methods section has to be written in future tense.
- Note: Page numbering starts (page number 1) on the first page of Chapter 1.

b. Literature Review

This chapter assesses literature in the field. It provides all background information. Unlike the Related Work chapter, it doesn't review individual papers. Instead, it gives a general background regarding the area of study as well as some background information about the tools to be used. Proper citation or referencing is very important in this chapter.

c. Related Work

This chapter reviews published papers in the area of the research. Each paper is reviewed in one or two paragraphs by presenting the approach, data sets used if any, results obtained, weaknesses, strengths, and lessons learnt. As a review, it should be short and brief and a review of a single paper should not span several pages.

While the "Literature Review" can reference published papers, books, Web sources, etc. the "Related Work" chapter should review only published papers in journals or peer-reviewed conference proceedings, master theses, PhD dissertations, book chapters (where a book is written by combining several papers as chapters) and technical reports.

For clarity of presentation, the chapter can be organized into sections so that each section can include papers dealing on the same subject. However, in order to break the chapter into sections, each section must review at least two papers.

Sometimes it may be difficult to find sufficient number of papers directly related to the area of research. In such situations, papers that are similar to the area of research can be reviewed. For instance, if the thesis is in text summarization for Amharic, papers in text summarization for other languages can be reviewed.

The chapter should be concluded by a summary section that outlines the gaps identified in those works and indicates those that the research will bridge or address.

d. The Proposed Solution

Please note that the titles and number of chapters from this point onwards may vary from thesis to thesis. Hence, this section should be considered as a general guideline and should be tailored towards the nature of each thesis.

The proposed solution usually contains a system architecture or a system model or a framework. It is the conceptual model that defines the structure and behavior of a system. It is a description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system.

A system architecture can comprise system components and the interaction and data flow among those components by means of uni- or bi-directional arrows. The modeling and management of persistent data (if any) is also described here. Use the proper symbols for input, output, process, database, etc.

Each component of an architecture should be discussed separately by making reference to the figure that shows the architecture.

- Note: Please be consistent in using names of components throughout the thesis. For instance, some use “pre-processor” and "preprocessor" interchangeably and lacks consistency. Use one of them throughout the thesis.

If an architecture or a model is taken from another source and modified (or adapted), then the modified part has to be clearly shown by way of shading or a box.

This chapter should clearly show "how" things are done. It is not sufficient to discuss "what" a given component does. Include algorithms and pseudo code to show how something is done. Clearly indicate what is taken from another source and what is modified or adapted as a result of the research, i.e., clearly show the contributions of your work. An algorithm should show the input, the process and the output. Don't use line numbers in algorithms unless you will make reference to some lines within the algorithm. Use proper indentation to clearly show the flow of control just as is done in writing programs using high level programming languages. All algorithms must be numbered and labeled at the bottom of the algorithm and must be referred at least once. The numbering follows the same guidelines for tables and figures given in the next Section.

Don't cover (repeat) issues that are addressed in previous chapters unless it is absolutely necessary to clarify some ideas.

If you use sequence diagrams (which is usually the case in projects), include few (say 2 or 3) in this chapter and annex the rest.

For a project, this chapter will have a title "System Analysis" and shall include issues such as functional and non-functional requirements and system modeling such as using use cases, class and sequence diagrams and an activity diagram. Then it will be followed by a "System Design" chapter and covers issues such as design goals, system architecture, sub-system decomposition, hardware/software mapping, and persistent data storage and management.

e. Experimentation/Prototype and Evaluation

What is proposed as a solution must be justified by way of a prototype or simulation at least partially. Otherwise it becomes only a wish.

Describe very briefly the tools and programming languages used in the experiment and why they were selected. Also describe the experimental setting such as capacity of the computer, operating system, etc. Then, document the experimental procedure and the results obtained.

If there are similar works done before, evaluate your work by comparing it with those previous works. The evaluation metrics vary based on the nature of the research. An example is comparing performance in terms of precision and recall. If there are no similar works then document the results by quantifying the various measurement metrics so that it can serve as a benchmark for future works in the area.

For a project, this chapter will have a title "Implementation" and documents the implementation with sufficient and selected screenshots from the prototype. A last section in this chapter has to be dedicated to user acceptance testing. The aim is to show that the developed system is useful and acceptable by end users. Prepare a questionnaire and distribute it to potential end users together with the developed system so that they can provide their ratings regarding the functionalities of the system, user friendliness, etc. Then compile, analyze and present the results.

A users' manual shall also be prepared for projects which shall be attached with the main report as a separate document.

f. **Conclusions, Recommendations (if any), and Future Work (optional)**

This chapter usually (but not always) has three parts. First, provide a summary of the thesis by outlining the activities done. Indicate how the general and specific objectives of the thesis are addressed and how the research questions are answered. No new material should be included here. The second part provides a bulleted list of contributions of the work. Contributions should not be a list of activities done rather it should document the achievements of the research. Finally, indicate future works to extend the work which were not done in the research due to several factors such as limitations in time, resources, etc. Note that recommendations can also be forwarded as long as they can be inferred from what is written in the thesis. These are issues that require policy decisions and other administrative actions that are beyond the capacity of the author.

- Note: The use of references is unusual in this chapter.

4.3 Tables, Figures, and Equations

Each figure and table must be numbered separately and labeled.

- Don't use a full stop at the end of a label.

The table/figure is numbered using section number, a full stop, and serial number, such as 2.1, 3.4, etc. That means the numbering is reset for every chapter. The labels should be informative. Use a font style for labels that is different from the body text which may vary depending on the taste of the author. Pick a style that you think is best from books or papers.

For equations start from 1 and continue numbering throughout the thesis. Use normal brackets and put the equation number right justified in front of the equation. Don't use square brackets since it will be confused with citations. After an equation, use a "where" clause and state what each of the components (variables, constants, etc.) of the equation are (or stand for). The following is an example of using an equation.

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N} \tag{1}$$

where μ is the population mean, N is the number of elements of the population, and X_i are observed values of elements of the population.

A table, a figure or an equation that is taken from a different source requires a citation. Put the citation in the body of the thesis when the figure, the table or the equation is referred and not on the labels. Alternatively, the source can be indicated at the bottom of the table or the figure separate from the label as a caption. This will also have an advantage of not including the citation in the "List of Tables" or "List of Figures" section since they are automatically generated. See the following example on citations of tables from a different source (which also applies for figures).

The following are examples on how to refer to tables and figures.

Table 5.1 [23] shows so and so. or As shown in Table 5.1 [23], [put here any conclusions]. Figure 5.1 shows so and so. or As shown in Figure 5.1, [put here any conclusions].

- Note: The first letter of the terms "Figure and "Table" has to be in capital letter even when used at the middle of a sentence.

The position of labels must be below the figure for figures and for tables above the table and both must be centered. Figures and tables must also be centered. Keep the label on the same page as the figure or table.

Each figure and table must be referred in the body of the thesis at least once. Remember that the reader will check tables and figures only when they are referred. Tables and Figures can appear before or after they are referred but should not be far away from their first referencing. Some word processors are not controllable and may insert a table or a figure anywhere before or after the referencing. Hence, don't make references to tables and figures using words such as "above" or "below". Don't refer to figures and tables as "… as shown in the following/above table/figure …". Instead explicitly refer to the table/figure using their numbers as "… as shown in Table/Figure 3.1 …". The above discussion also applies for equations.

Overcrowding tables with lines should be avoided. Horizontal lines should be included only for the heading and at the end of the table. If there is no row of totals or something similar, use only a single line for the bottom of the last row. The following examples demonstrate the various issues discussed so far. If a table spans more than one page "Repeat Heading Rows".

Table 5.1: *List of Courses to be offered in the first Semester*

Course Code	*Course Title*	*Credit Hours*
COSC 601	Research Methodology	3
COSC 664	Advanced Computer Networks	4
COSC 621	Distributed Systems	4
Total		*11*

Table 5.2: *Nationally set Annual intake Targets from 2001 to 2005*

Year / *Intake*	*2001*	*2002*	*2003*	*2003*	*2005*
Annual Intake - ICT	6, 000	9, 000	11, 000	11, 000	11, 000
Annual Intake - All	90, 000	110, 000	110, 000	110, 000	110, 000
% ICT Intake	*6.67*	*8.18*	*10*	*10*	*10*

Figure 5.1: *Emblem of AAU*

Tables and figures must not extend beyond the margins.

It is a common mistake to insert figures that are generated by a software (which is usually the case in simulation experiments) without knowing how they are interpreted. This has to be avoided. Use only those figures for which proper interpretation can be made.

Overcrowding the thesis with so many figures and tables is not recommended. Use few of them in the thesis and the rest can be included as annexes.

If figures obtained from other sources do not print well, try to draw them yourselves if possible. This also applies to tables and equations.

Tables are used if there are more than two rows and columns. If a table contains very few entries, it is better to avoid using it and present the information in textual form.

If you create a figure such as a line graph to represent quantitative values, properly label the X- and Y-axes. Don't use titles in such figures since they will be similar to the figure labels and become redundant. Use proper charts and graphs. For instance, use line graphs to show trend, i.e., whether a dependent variable will increase or decrease as the independent variable increases or decreases.

4.4 References

Different authors use different styles for citation. The commonly used style in the Science and Engineering fields is the one by the Institute of Electrical and Electronics Engineers (IEEE) where a citation number is enclosed within square brackets and the reference list is arranged by the order of citation in the thesis, not by alphabetical order.

Every reference cited in the text must appear in the reference list (usually titled "References"). By the same token, all entries in the reference list must have been cited in the text at least once. The following are examples of citations.

The above sentence may be true as pointed out in [2].

- Note: If the citation is at the end of a sentence, the full stop appears after the closing square bracket.

Rabinovich and Spatscheck [1] concluded that …

As presented in Sen *et al.* [4], a mobile entity is … (Use *et al.* if the authors are more than two). Note* that "et al." is a scholarly abbreviation of the Latin phrase et alia, which means "and others." It is commonly used when one doesn't want to name all the people or things in a list, and works in roughly the same way as "etc."

- Note: *et al.* can not be used in the "References" section. Instead list all authors since it is an acknowledgement of their co-authorship.

The authors in [6] argue that …

- Note: it is recommended to use the word "author" instead of other alternatives such as "scholar", "researcher", "academician", etc.

Misganaw Kebede [5] pointed out that …

Use family name for non-Ethiopian authors and full name for Ethiopian authors, as shown in the previous examples. Don't use "he" or "she" to refer to an author even if you know the sex of the author (which can mostly be inferred from the name for Ethiopian authors). However, it is not always possible to infer the sex of the author from the name. Instead, use terms such as "the author". Don't use the title† of authors such as "Dr.", "Professor".

* Source: https://public.wsu.edu/

† An exception is in writing acknowledgements where titles can be used.

Don’t mention names of people who provided information; but a list of interviewees may be included at the end of the report or can be acknowledged in the acknowledgement page.

Don’t put too many references at once such as [2, 12, 14, 16, 18, 19, 33, 43] since it will be difficult for the reader to find out which information is taken from which source. Unless there are exceptional cases, don’t use more than three references at once. When there are more than one reference, write them in ascending order and within 2 square brackets such as [3, 6, 10] and not as [10, 3, 6] or [3], [6], [10]. Use a single space after the comma.

References must be listed in the “References” section in the order they are used in the thesis.

All factual information shall be substantiated with referencing. However, referencing should not be an excuse for not defending a referred idea. You have to include ideas only if you are convinced that it is correct and that you can defend it. Try to use recent references. For instance, the following doesn’t make sense.

“Ethiopia is a large country with a population of 74 million [8]”.

The above sentence may be true at the time reference [8] is written but doesn’t hold true in 2015, the year the thesis is written. In such cases either search for recent information or clearly indicate the date of the reference.

References have two major objectives. Firstly, the thesis author is acknowledging the works of others thereby avoiding plagiarism. Secondly, readers who need more information can access the referred material. Hence, all references must be traceable. Formats vary, but an entry for a book usually contains the following information to be traceable:

- author(s)
- title
- publisher
- date of publication

An entry for a journal or a conference proceeding paper usually contains:

- author(s)
- article title
- journal title or conference details
- volume and number
- pages on which the paper is printed in the journal or conference proceeding
- date of publication

The following are example styles for a book, a journal paper, a conference proceeding, and a Web reference, respectively. Please note the double quotes, italicization and how Vol., No., and pp. are written. Also use hanging indent as shown in the following examples.

[1] J. Watkinson, *The MPEG Handbook*, Focal Press, Oxford, 2001.

[2] R. J. Flynn and W. H. Tetzlaff, "Multimedia - An Introduction," *IBM Journal of Research and Development*, Vol. 42, No. 2, 1998, pp. 165-176.

[3] C.-H. Chi, Y. Cao, and T. Luo, “Scalable Multimedia Content Delivery on Internet,” in *Proceedings of the IEEE International Conference on Multimedia*, Lusanne, Switzerland, August 2002.

For information obtained from the Web, include the URL and the date when the site was last accessed. The word processor may underline or color such sources to show that it is a link for quick navigation. Please delete the link so that it will not be underlined or colored. The following is an example of a reference from the Web.

[4] Zona Research, “The Economic Impacts of Unacceptable Web Site Download Speeds,” white paper, 1999, retrieved from http://also.co.uk/docs/speed.pdf, Last accessed on June 10, 2014.

As much as possible avoid using Web sources that are not peer-reviewed such as Wikipedia as justifications of arguments. Such sources may be good for understanding ideas but are not peer-reviewed and hence some of the information may be “own” view of the author(s) which is sometimes controversial.

If a masters thesis or a PhD dissertation is referenced, then use “Unpublished” as in the following example. For Ethiopian authors use the full name of the author.

[5] Misganaw Kebede, “QoS Aware Routing Protocol for MANETs”, Unpublished Masters Thesis, Department of Computer Science, Addis Ababa University, 2013.

Since Computer Science is a dynamic field, try to use recent references. Except for some factual information, the results and conclusions of most papers in Computer Science become outdated in few years.

4.5 Statistical Analysis

If any statistical analyses are made it is better to get the advice of a statistician. Statistics is a science that plays a significant role in analyzing, interpreting and presenting data. It dictates which statistical method can be used with which type of data. For instance, the arithmetic mean cannot be used as a measure of central tendency for ordinal data (such as Excellent, Very Good, Good, Poor, Very Poor) even if they are coded (as 5, 4, 3, 2, 1). The coding is for convenience and the numbers do not mean anything. For instance, what is the average of Excellent and Very Good? Of course, $(5+4)/2 = 4.5$. If this is translated to (Excellent + Very Good)/2, no meaning can be attached to it. For such data the appropriate measures of central tendency are mode or median.

If a sample survey is conducted, identify the sample and the population. A sample has to be a representative of the population. Hence, deciding the sample size and selecting the samples have to be done properly following the right statistical method. If a questionnaire is used, prepare the questions carefully. Avoid trivial questions and each question should be used in the analysis later. The questionnaire has to be annexed. Attach a blank questionnaire. Do not attach a completed questionnaire. If the questionnaire is prepared in a language other than English, both the original version and the translation to English have to be annexed.

If interviews and focus group discussions are conducted, then an interview guide and a focus group discussion guide have also to be prepared and annexed.

The results obtained from the sample survey and/or the interviews and/or the focus group discussion have to be properly reported in the body of the thesis. If such results are used for proposing a solution, this has to be clearly indicated.

4.6 Typing Considerations

- The thesis must be typed on A4-size paper, preferably on one side of the paper only.
- Font and font size: One of the following fonts should be used for all text except algorithms and pseudo codes: Times or Times New Roman. Use `Courier` (or `Courier New`) for algorithms and pseudo codes. Use a 12-point font size. Use bigger fonts (incrementing by 1) for titles and subtitles depending on the number of labels (depth or hierarchy) of titles.
- There must be a margin of 1.3 inch on the left hand side of the page to allow for binding, and margins of 1 inch for the top, right, and bottom.
- The typing must be 1.5-spaced.
- Don't use a blank paragraph to create spacing between paragraphs. Instead use 6 points below and 0 points above a paragraph.
- All paragraphs (except those indicated in the respective sections) should be justified, i.e., they are not ragged on the right margin similar to the paragraph you are reading now.
- All pages shall be numbered consecutively in the bottom, center position. The pages must be numbered in Arabic numerals starting from the first page of the Introduction chapter as page 1. Previous pages starting from the Table of Contents page must be numbered

using lower case roman numerals. All pages prior to the Table of Contents page do not have page numbers.

- The thesis should not exceed 100 pages excluding references and annexes (if any). Remember that the quality of a thesis is not judged by the number of pages. Be focused and write to the point.

4.7 General Considerations

- In writing a thesis repetitions have to be avoided as much as possible. It makes the thesis unreadable and boring if sentences or paragraphs are repeated here and there throughout the thesis. This leads to what is known as "verbose" writing.
- The thesis should be understandable. If terms that are not yet covered are used, give a cross reference so that the reader can refer to those terms. Sections must be coherent and connected with each other.
- Never use I or my. Instead use We or our. Avoid the use of "you" and its derivatives which is generally the case in writing a user manual for a software. For instance, we have used such kind of writing in this guideline, but not recommended for a thesis.
- Don't start sentences with words such as And. Don't also start a sentence with references. The following is a bad style. "[5] defines research as …".
- Know the difference between Internet and internet. While internet (with lowercase i) refers to a network of two or more networks (also known as an internetwork), Internet (with uppercase I) refers to a collaboration of more than hundreds of thousands of interconnected networks worldwide.

- Know the difference between the Internet and the Web. The Internet is a network of networks that connects millions of computers together globally, forming a network in which any computer can communicate with any other computer as long as they are both connected to the Internet. The World Wide Web, or simply Web, is a way of accessing information over the Internet as a medium. It is an information-sharing model that is built on top of the Internet. The Internet is used not only to have access to information on the Web, but also for other applications such as e-mail, file transfer, remote login, etc.
- Avoid copy/paste. This can be easily detected; for instance, use of rare and unusual terms as is usually the case with non-English speakers through the use of Thesaurus. Take ideas and use your own English.
- Avoid advert like statements. This is mostly the case in commercial Web pages. Statements such as "xyz is the most widely used word processing software in the world" is an advert like statement. Such statements can be used only if supported by proper evidence (citation) which has to be the result of a study instead of a statement by the manufacturer.
- Proofread the thesis many times before submission. We usually overlook our own errors. Hence and if possible, give it to a friend even if s/he is not in the field.
- Use the feature of the word processor to correct common spelling and grammatical errors. However, be careful since it doesn't mean that all the suggestions of a word processor are correct.

- The thesis has to be free of grammatical and typographic errors as far as possible. The major error that is most observed is subject-verb disagreement. The subject of a sentence must agree with the verb of the sentence: in number: singular vs. plural; in person: first, second, or third person. Hence, it has to be reviewed by a language expert to avoid such errors. Remember that the thesis will be archived and read later by your students. In such situations, such errors will be an embarrassment to you. What is more annoying is when the thesis is a mix of paragraphs with no errors and others full of errors which is usually an indication of copy/paste. Advisors are not also happy to have their names appear in a thesis if it is badly written.
- One problem that is often observed is incomplete sentences. The following is an example. "Since the secret key must be transferred using secure channels." The previous sentence is incomplete and doesn't convey any meaning.
- Use the SI (International System of Units) standard for measuring length, weight, etc. SI uses meter and its derivatives (instead of inch, feet, mile or others) for length and kilogram and its derivatives (instead of pound and others) for weight. An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive". Use a space between the number and the unit symbol such as 5 kg instead of 5kg.
- Use a zero before decimal points such as "0.25" and not ".25".
- Don't write words in all capital letters unless it is an acronym or an abbreviation. The first letter of proper names such as authors, programming languages, etc. has to be in capital.

- Avoid using colors if the thesis is not to be printed in color, which is not usually a requirement for a scientific writing. For figures, it is better to use proper shading instead of colors. Note that sometimes color may not be avoidable such as in image processing or in taking screenshots. In this, case make sure that each color is identifiable in a grey scale print out or selectively print those pages in color.
- The abbreviation “i.e.,” means “that is”, and the abbreviation “e.g.,” means “for example”. In both cases please note the comma after the full stop.
- Use a space before (not after) open bracket. Don’t use a space before close bracket.
- If you are referring to a chapter or a section in the thesis you are writing, capitalize the first letter of the words chapter and section. For instance, you could write “As discussed in **C**hapter **T**wo, …” or as indicated in **S**ection 3.2, …”, etc. Small letters have to be used in references such as “in the previous chapter” or “in the next section”, etc.
- Don’t use phrases such as “In the last 10 years …” since this is relative to when the thesis was written. Instead write as “As of the year 2005 …”.
- Avoid using terms such as “ … in this country …” or “ … in our country …” although you may be referring to Ethiopia. Remember that the thesis may be read by international readers. Hence, write explicitly as “ … in Ethiopia …”.

- If you use text boxes, make sure that the text fits into the box. It is mostly observed that half of the line is displayed while the remaining half is hidden or some text is totally hidden.
- Use a space after a full stop that ends a sentence and before the first character of the next sentence. Don't use a space before a full stop, a comma, a semicolon, a colon or a question mark. Use a space after each of these characters.
- If you are using non-English characters such as Ethiopic characters, make sure that the printout is readable when printed from another computer. Converting to a PDF format and printing the PDF version may solve most of such problems.
- Don't use headers and footers (except page numbers as footers).
- Note that there is a confusion in the use of some measurement units. While network bandwidth is measured in Kbps (Kilobits per second) or Mbps (Megabits per second) or Gbps (Gigabits per second), memory and disk capacity are measured in KB (Kilobytes) or MB (Megabytes) or GB (Gigabytes). For instance, we write 10 Mbps (not 10 MBps) for bandwidth and 10 MB for disk space.
- Use hanging indent for bulleted and numbered lists as the paragraph you are reading now.
- Take care not to write compound words as two separate words. Hence, write as "framework" not "frame work", "overview" not "over view", "keyword" not "key word", "database" not "data base", etc.
- Note that some terms such as "equipment", "hardware", and "software" do not have plural forms.

5. Roles and Responsibilities

This Section outlines the roles and responsibilities of the advisor and the student in carrying out a thesis. It has to be emphasized that an advisor has only an advisory role and as such the responsibility of carrying out all activities required for a thesis towards its completion is that of the student. Taking all initiatives such as meetings with the advisor, presentations, submitting papers to conferences, workshops, journals, etc. are also the responsibilities of the student.

5.1 Roles and Responsibilities of Advisors

A thesis is undertaken under the guidance of a thesis advisor who is selected by the student and approved by the DGC. The advisor should have the expertise in the area of the thesis research with a minimum academic rank of assistant professor.

The role of the advisor is to work with the student throughout the entire process, including selection and determination of procedures, and preparation of the final product, in addition to helping ensure all requirements are met.

It is important that the student maintains communication with his/her thesis advisor throughout the work. The advisor should be available to meet regularly with the student, to evaluate the progress of his/her research, to discuss the problems that might arise, and to provide whatever encouragement or direction that is deemed necessary. These meetings should be regular to review the student's progress in writing the thesis.

It is the advisor's responsibility to approve the student's thesis as complete before it is given to the Examining Committee.

In general, an advisor has the following roles and responsibilities:

- To guide the thesis activity: the advisor shall provide appropriate guidance on the nature of research, the expected standards, and about how to plan and conduct the research so as to ensure that the normal expectation of submission will be met.
- To be available for consultation at relatively short notice: an advisor shall maintain regular contact with his/her student through meetings according to mutually agreed frequency and duration. Normally, a minimum of one hour per week (on average) shall be set aside for consultation. All communications should not necessarily be face to face; other electronic means can also be used as alternative means of communication.
- To be as helpful as possible in suggesting the next course of action and in assisting the student in carrying out the research.
- To inform/guide the student approximately how long it will be before written work, such as drafts of chapters/sections, can be returned with comments.
- To guide the student on the use of original literature and sources and shall give guidance on avoiding plagiarism.
- To be thorough in the review of thesis chapters/sections, supplying, where appropriate, detailed comments on such matters as literary form, structure, use of evidence, relation of the thesis to published works on the subject, footnoting, and referencing techniques, and making constructive suggestions for rewriting and improving the draft.
- To encourage and guide the student to publish the results in journals and/or present in conferences and workshops.

- To indicate clearly when a draft is in a satisfactory final form or, if it is clear to the advisor that the thesis cannot be successfully completed, to advise the student accordingly.
- To advise and help the student to approach other faculty members for assistance with specific problems that are beyond the capacity (expertise) of the advisor or even to request the reading of a section of the thesis.
- To know and understand that the ownership and the right of distribution of all software and other products that are developed as a result of a thesis work belong to the University.
- To report to the DGC on the progress of the work.
- To comply with all ethical requirements.

5.2 Roles and Responsibilities of Students

When a student undertakes a thesis, s/he assumes several roles and responsibilities, the most important of which are listed below.

- To choose a topic (often with the advisor's help) and to produce a thesis that is essentially his/her own work. The thesis should meet the standards of scholarship including demonstration of the capacity for independent scholarship and research in the field.
- To acknowledge direct assistance or borrowed material from other researchers which otherwise is considered as plagiarism even if unintentional. They should make themselves aware of rules and regulations on proper citation, including copyright and intellectual property regulations to avoid representing as their own work the work of another as this will result in a severe charge of plagiarism.

- To realize that the advisor has other duties which may at times delay the student's access to the advisor at short notice.
- To give serious and considered attention to the advice and direction from the advisor.
- In situations where any form of publication is produced from the thesis work, the authorship has to be jointly with the advisor.
- To know and understand that the ownership and the right of distribution of all software and other products that are developed as a result of a thesis work belong to the University.
- To understand clearly the requirements of the program.
- To know the University's regulations and standards to which the writer of a thesis is required to conform.
- To submit the thesis on time. A thesis shall be submitted as early as possible so that there is sufficient time for examiners to read the thesis and also to incorporate comments after the defense.
- To know that the advisor may cancel advisement under certain circumstances including but not limited to the following: if the student doesn't demonstrate a progress in his/her work, or is not in a regular contact with the advisor, or is not following the advises given by the advisor, etc.
- To comply with all ethical requirements.

6. Thesis Submission and Defense

6.1 Submission of Thesis

The format of the thesis shall be as that provided in Annex C - Thesis Template. A thesis shall be submitted to the Department in three bound copies at least one month before the date of defense, unless and otherwise a special consideration or decision is made by the DGC. If there is a substantial change in the thesis after it is submitted and before the defense, the changed pages must be submitted to the advisor in three copies so that the advisor can distribute it to the examiners ahead of time.

6.2 Thesis Defense

When a student, after conferring with the advisor, submits his/her thesis, the DGC appoints an Examining Committee. The Committee consists of two examiners and the advisor. The advisor is a non-voting member. The date and venue of the defense must be posted at least one week prior to the defense date. The thesis must also be distributed to the examiners and the advisor at least two weeks prior to the defense date.

The defense is conducted as outlined below.

- The student shall defend his/her thesis before the Examining Committee. The defense session is open to anyone who is interested to attend.
- The thesis defense is presided over by an impartial person appointed by the Department as chairperson.
- The student first presents his/her thesis orally with the aid of a PowerPoint presentation to the Examining Committee.
- After the defense, which includes a maximum of thirty minutes for presentation and sixty minutes for cross-examination by the two examiners, the decision of the Examining Committee may fall under the following categories. These decisions shall not be made in the presence of the advisor.

- **Accepted**: The thesis is accepted with no change or with only minor changes that do not require the advisor's approval.
- **Accepted with minor modifications**: Corrections can be made immediately and to the satisfaction of the advisor.
- **Accepted with major modifications**:
  - The thesis requires substantial changes, which are to be made to the satisfaction of the Examining Committee or its designate. It is not necessary for the Examining Committee to reconvene.
  - If the required changes are not made to the satisfaction of the Examining Committee or its designate, the thesis is automatically considered as rejected.
- **Rejected**: A thesis will be rejected if:
  - The work is found by the Examining Committee not to have met the required standards; or
  - The work is plagiarized as judged by the Examining Committee; or
  - The thesis was accepted with major modification but the required changes are not made to the satisfaction of the Examining Committee or its designate; or
  - The work has already been used to confer a degree from this or another institution. However, this shall not preclude the student from submitting such work provided enough extra work has been carried out to expand the scope and depth of the subject.

Thesis Rating

- A thesis that is defended and accepted or accepted with minor modifications or accepted with major modifications shall be rated as "Excellent", "Very Good", "Good" or "Satisfactory".
- A rejected thesis shall be rated "Fail".

Final Submission - After Thesis Defense

A student who successfully defended his/her thesis shall submit the items listed in Annex D - Guideline for Submitting Final Thesis - to the Department. The final version of the thesis must include any required modifications given by the Examining Committee.

Upon final submission of the thesis, a student shall be deemed to have granted the University a non-exclusive, royalty free license to reproduce, archive, preserve, and communicate to the public by any means including the Internet, loan, and distribute the thesis worldwide for non-commercial purposes, in any format.

6.3 Tips on Slide Preparation and Presentation

Students shall prepare a PowerPoint presentation and present their work. The following are some tips regarding slide preparation and presentation. Note that the guideline in this Section also applies for thesis proposal defense.

a. Slide Preparation

- Slides must be numbered.
- Use bigger fonts (at least 22 point) for readability.
- Don’t overcrowd the slides with text; be selective.

- Choose a plain background that is simple and makes the text readable.
- Don't include new material that was not covered in the thesis although additional diagrams, illustrations, and examples are acceptable.
- Use effects when absolutely necessary.
- Show it to your advisor for comments and suggestions of improvement.

b. Presentation

- During presentation, bring a copy of the thesis since mostly questions refer to the thesis. Also bring paper and pen so that you can write comments for improvement.
- Usually you will be given 30 minutes for presentation. Hence, rehearse it many times so that you can finish within the allotted time. Let a friend in the audience indicate to you the time left with what ever sign the two of you agree.
- Dress neatly.
- Be confident - read and understand the material beforehand and try to present without reading. Note that the whole purpose is to convince the examiners that you have mastered the subject matter.
- Be neither very fast nor very slow (this is very subjective).
- Face participants; don't be shy. This shows your confidence towards the subject.
- Speak loudly so that the audience at the far end of the hall can follow your presentation.

- During the question and answer session:
  - Address questions properly.
  - Be ready to be challenged; don’t be offended; it is the way of life in academics. Examiners may ask questions to make sure that you very well know what you are presenting.
  - Say I don’t know if you don’t know the answer.
  - If there are certain issues for which there was an agreement between you and your advisor, explain this in your answer.

Annexes

Annex A - Thesis Proposal Template

Identification (on cover page)

Addis Ababa University

Masters Thesis Proposal

Name of Student: ______________________________________

College: _______Natural Sciences_________________________

Department: ____Computer Science_______________________

Major: ________Computer Science_________________________

Name of Advisor: _______________________________________

Title of Thesis: ____

Inner Pages

Table of Contents (on a separate page)

1. Introduction
2. Motivation
3. Statement of the Problem
4. Objectives
5. Methods
6. Related Work
7. Scope and Limitations
8. Application of Results

Annexes

Annex A: Timetable

Annex B: Cost

References

Last Page

Submitted by:

Student	Signature	Date

Approved by:

1. Advisor — Signature — Date
2. Chairman, Department's Graduate Committee — Signature — Date

Annex B - Thesis Proposal Defense Result Submission Form

Addis Ababa University

College of Natural Sciences

Department of Computer Science

Name of Student: ______________________________________

Title of Thesis: ____

Decision of the Examining Committee, please tick.

- ☐ Approved
- ☐ Clarification Required
- ☐ Revise and Resubmit
- ☐ Rejected

If the decision is other than “Approved”, please provide the details below indicating why the proposal is not approved. You can attach your comments on separate sheets of paper if extra space is required.

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1. ______________ ______________ ______________

   Reviewer Signature Date

2. ______________ ______________ ______________

   Advisor Signature Date

Annex C - Thesis Template

The organization depends on the nature and material contained. However, the following general format is suggested.

- **Preliminaries**
    - Title Page
    - Signature Page
    - Abstract
    - Dedication (if any)
    - Acknowledgments
    - Table of Contents
    - Separate List of Tables, Figures, Algorithms, etc., if any
    - Acronyms and Abbreviations (if any)
- **Body of the Thesis**
    1. Introduction
        - 1.1 Background
        - 1.2 Motivation
        - 1.3 Statement of the Problem
        - 1.4 Objectives
        - 1.5 Methods
        - 1.6 Scope and Limitations
        - 1.7 Application of Results
        - 1.8 Organization of the Rest of the Thesis
    2. Literature Review
    3. Related Work
    4. The Proposed Solution

5. Experimentation/Prototype and Results
6. Discussion
7. Conclusions, Recommendations (if any), and Future Work (optional)

- References (no chapter/section number is given)
- Annexes (if any) (no chapter/section number is given)
- Signed Declaration Sheet

Annex D - Guideline for Submitting Final Thesis

A student who successfully defended his/her thesis should submit the following to the Department Secretary. No services will be provided by the Department before submitting these requirements.

1. Three bound copies of the thesis after incorporating the comments given by the Examining Committee during defense and signed by all Examining Committee members.
2. The following electronic copies on a CD:
    a. Soft copy of the thesis (both in MS Word and PDF format),
    b. Abstract of the thesis (both in PDF and MS Word format) as separate files,
    c. Users' manual of the prototype system developed (if any),
    d. Source code of the developed system (if any),
    e. Sample data used in the system or database (if any),
    f. Relevant electronic reference materials that were used during the work (under a directory) sorted as they appear in the "References" section; and
    g. Tools and systems used in the work.

Please verify with your advisor the completeness of the documents before submission.